JAVA ENTERPRISE PRACTICALS

Practical 1: Simple Calculator Using Servlet

Environment: Apache NetBeans · GlassFish Server · JDK 21

1. Aim

To develop a simple calculator web application using Java Servlet to perform addition, subtraction, multiplication, and division of two numbers.

2. Software Requirements

- Java JDK 21
- Apache NetBeans IDE
- GlassFish Server
- Web browser
- HTML and Java

3. Theory

A Servlet is a Java server-side component that processes client requests and generates responses. In this practical, an HTML form accepts two numbers and an arithmetic operation. The form sends the data to a Java Servlet, which performs the calculation and displays the result in the browser.

4. Algorithm

1. Create an HTML form to accept two numbers.
2. Provide options for addition, subtraction, multiplication, and division.
3. Submit the form to the Servlet using HTTP POST.
4. Retrieve the input values using `request.getParameter()`.
5. Convert the input strings to numbers using `Double.parseDouble()`.
6. Use a switch statement to perform the selected operation.
7. Display the result using `PrintWriter`.
8. Handle invalid input and division by zero.

5. Project Creation in NetBeans

1. Open NetBeans.
2. Select **File** → **New Project**.
3. Select **Java Web** → **Web Application**.
4. Enter project name: **SimpleCalculator**.
5. Select GlassFish Server.
6. Choose the Java EE/Jakarta EE version supported by the installed server.
7. Click **Finish**.
8. Under Web Pages, create `index.html`.
9. Under Source Packages, create package `com.calculator`.
10. Inside that package, create Servlet `CalculatorServlet`.

6. Program 1: index.html

Save the following code in `index.html` under Web Pages.

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple Calculator</title>
</head>
<body>
    <h2>Simple Calculator Using Servlet</h2>

    <form action="calculator" method="post">

        <label>Enter First Number:</label>
        <input type="number" name="num1"
               step="any" required>
        <br><br>

        <label>Enter Second Number:</label>
        <input type="number" name="num2"
               step="any" required>
        <br><br>

        <label>Select Operation:</label>
        <select name="operation">
            <option value="add">Addition (+)</option>
            <option value="sub">Subtraction (-)</option>
            <option value="mul">Multiplication (*)</option>
            <option value="div">Division (/)</option>
        </select>
        <br><br>

        <input type="submit" value="Calculate">

    </form>
</body>
</html>
```

Explanation:

- `action="calculator"` specifies the Servlet URL.
- `method="post"` submits the form using HTTP POST.
- `name="num1"` and `name="num2"` identify the number fields.
- `name="operation"` identifies the selected operation.
- `required` prevents empty input fields from being submitted.

7. Program 2: CalculatorServlet.java

Create the Servlet in package `com.calculator`.

Important: The code below uses `jakarta.servlet`, suitable for GlassFish 6 or later. GlassFish 5 requires the equivalent `javax.servlet` imports.

```
package com.calculator;

import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/calculator")
public class CalculatorServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
```

```
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        PrintWriter out = response.getWriter();

        try {
            double num1 = Double.parseDouble(
                    request.getParameter("num1"));

            double num2 = Double.parseDouble(
                    request.getParameter("num2"));

            String operation =
                    request.getParameter("operation");

            double result;

            switch (operation) {

                case "add":
                    result = num1 + num2;
                    break;

                case "sub":
                    result = num1 - num2;
                    break;

                case "mul":
                    result = num1 * num2;
                    break;

                case "div":
                    if (num2 == 0) {
                        out.println(
                            "<h3>Division by zero is not allowed.</h3>");
                        return;
                    }
                    result = num1 / num2;
                    break;

                default:
                    out.println("<h3>Invalid operation!</h3>");
                    return;
            }

            out.println("<!DOCTYPE html>");
            out.println("<html><body>");
            out.println("<h2>Calculator Result</h2>");
            out.println("<p>First Number: " + num1 + "</p>");
            out.println("<p>Second Number: " + num2 + "</p>");
            out.println("<p>Operation: " + operation + "</p>");
            out.println("<h3>Result: " + result + "</h3>");
            out.println("<a href='index.html'>Go Back</a>");
            out.println("</body></html>");

        } catch (NumberFormatException e) {
            out.println("<h3>Please enter valid numbers.</h3>");
        }
    }
}
```

8. Important Code Explanation

Statement	Purpose
@WebServlet("/calculator")	Maps the Servlet to the /calculator URL.

Statement	Purpose
extends HttpServlet	Defines the class as an HTTP Servlet.
doPost()	Handles HTTP POST requests.
request.getParameter()	Retrieves submitted form values.
Double.parseDouble()	Converts strings into numbers.
switch	Selects the arithmetic operation.
response.setContentType()	Sets the response content type to HTML.
response.getWriter()	Obtains a writer to send output to the browser.

9. Execution Steps

1. Save both files using **Ctrl + S**.
2. Right-click the project and select **Clean and Build**.
3. Check the NetBeans Output window for build errors.
4. Right-click the project and select **Run**.
5. Wait for GlassFish to deploy the application.
6. Open the application's `index.html` page.
7. Enter two numbers.
8. Select an operation.
9. Click **Calculate**.
10. Verify the result in the browser.

The application URL commonly resembles:
`http://localhost:8080/SimpleCalculator/`
The port and context path may vary according to your server configuration.

10. Expected Output

For first number **20**, second number **5**, and **Addition**:

```
Calculator Result

First Number: 20.0
Second Number: 5.0
Operation: add
Result: 25.0

Go Back
```

11. Test Cases

First Number	Second Number	Operation	Result
20	5	Addition	25.0
20	5	Subtraction	15.0
20	5	Multiplication	100.0
20	5	Division	4.0
20	0	Division	Error message

12. Common Errors

Error: package jakarta.servlet does not exist

Check the GlassFish version and project server configuration. Make sure the correct Servlet API is available.

Error: HTTP 404

Check the Servlet URL mapping and the HTML form's action attribute.

Error: GlassFish fails to start

Verify that the server uses a compatible Java version. The Java platform must be registered and selected correctly in NetBeans.

Error: Build failed

Read the first compiler error in the Output window and correct the indicated problem.

Compatibility note: GlassFish 5 uses `javax.servlet`. GlassFish 6 and later use `jakarta.servlet`. Do not mix these namespaces.

13. Viva Questions and Answers

Q1. What is a Servlet?

A Servlet is a Java server-side component that processes client requests and generates responses.

Q2. What is GlassFish?

GlassFish is a Java application server that runs Java web applications.

Q3. What is the purpose of doPost()?

It handles HTTP POST requests.

Q4. What is HttpServletRequest?

It provides access to request data, including submitted form parameters.

Q5. What is HttpServletResponse?

It is used to configure and send the response to the client.

Q6. What is @WebServlet?

It maps a Servlet class to a URL pattern.

Q7. Why is getParameter() used?

It retrieves values submitted through HTML form fields.

Q8. Why is Double.parseDouble() used?

It converts input strings into numeric values for arithmetic calculations.

Q9. What is the difference between GET and POST?

GET commonly sends parameters in the URL, whereas POST sends submitted data in the request body.

Q10. What is PrintWriter?

It writes text or HTML to the HTTP response.

Q11. What happens when the denominator is zero?

The program displays an error message instead of performing the division.

Q12. What is the role of HTML in this practical?

HTML provides the form through which the user enters numbers and selects an operation.

14. Result

The Simple Calculator web application was developed using Java Servlet and HTML. It performs addition, subtraction, multiplication, and division and displays the results in a browser.

15. Quick Revision

Remember these five important points:

1. `@WebServlet("/calculator")` maps the Servlet URL.
2. `doPost()` processes the submitted form.
3. `request.getParameter()` reads the input values.
4. `switch` performs the selected arithmetic operation.
5. `PrintWriter` displays the result in the browser.